

Early Identification of At-Risk Learners from Digital Learning Traces

Comparative Learning Analytics using OULAD and Moodle Interaction Data

Team Member	Name	Roll Number	Contribution
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GITHUB LINK(code) : https://github.com/Prabhakar-Yadav/ET610_Assignment2

Member1: Completed the data processing, feature engineering, and modeling pipeline for the OULAD dataset. Extracted and integrated all seven relational tables, constructed temporal features for Day 30 VLE activity and Day 60 assessment performance, and trained the Random Forest classification model. Additionally, evaluated the model (achieving 0.8674 ROC-AUC and 0.79 accuracy) and generated all primary visual analytics, including the ROC curve, feature importances, and confusion matrix heatmap

Member2: Designed the Moodle LMS interaction framework and event-level clickstream parsing logic across quizzes, assignments, forums, and passive views. Conducted the reproducibility and execution diagnostic audit, synthesized pedagogical insights across both datasets, and translated model findings into practical course interventions. Furthermore, authored and structured the complete final project report, consolidating all hypotheses, methodology, and comparative findings.

Abstract

This assignment investigates whether early digital learning traces can identify students at risk of failure or withdrawal before major assessment deadlines. Two distinct educational contexts are studied: the Open University Learning Analytics Dataset (OULAD), which combines VLE clickstream, assessment, registration and demographic data, and a Moodle LMS interaction setting centered on event-level activity across quizzes, assignments, forums and passive resources. For each dataset, two domain-informed hypotheses connect self-regulation and active learning behavior to final outcomes. The executed OULAD notebook integrates 32,593 student registrations and 10.66 million VLE interaction rows, engineers early activity and assessment features, and trains a Random Forest model. The held-out test result is strong (ROC-AUC 0.8674; accuracy 0.79), with early assessment score, submission latency, assessment completion, early clicks and active-day regularity among the leading predictors. The Moodle technical analysis similarly frames active formative participation and access regularity as stronger early-warning signals than passive viewing. Together, the findings support a practical early-warning design that emphasizes what students do, how regularly they do it, and whether they complete early learning tasks on time.

1. Introduction & Hypotheses

Online learning systems continuously record behavioral traces, but instructors often see risk only after a student misses an assessment or receives a poor grade. Learning analytics can shift this process from retrospective diagnosis to early, explainable intervention. The study therefore asks whether temporally bounded engagement, assessment timing, active practice and social participation can provide useful warning signals before final outcomes are known.

Dataset	Hypothesis	Operationalization / expected evidence
OULAD - H1	Weak or declining early engagement is associated with later failure or withdrawal.	Early click volume and access regularity should carry predictive value beyond static demographics.
OULAD - H2	Late or weak early assessment behavior is associated with poorer final outcomes.	Early average score, assessment completion and submission latency should be strong predictors of risk.
Moodle - H1	Active formative interaction predicts success better than passive reading.	Quiz/assignment actions should contribute more than resource/page/url views.
Moodle - H2	Regular access and social participation are associated with retention.	Active-day count and forum participation should provide additional risk signal.

Research objective: Build and interpret early-warning models that go beyond descriptive visualization by combining feature engineering, temporal constraints and supervised machine learning.

2. Dataset & Preprocessing

2.1 OULAD data architecture

The OULAD pipeline uses all seven relational tables. The executed notebook confirms 32,593 student registrations, 173,912 assessment-submission records and 10,655,280 VLE interaction rows. The outcome is derived from studentInfo.final_result, while VLE and assessment tables provide early behavioral predictors.

Table	Executed shape	Role in analysis
assessments.csv	206 x 6	Assessment type, due date and weight
courses.csv	22 x 3	Module/presentation structure
studentAssessment.csv	173,912 x 5	Submission dates and scores
studentInfo.csv	32,593 x 12	Demographics, credits, final result
studentRegistration.csv	32,593 x 5	Registration/unregistration timing
studentVle.csv	10,655,280 x 6	Fine-grained VLE clickstream
vle.csv	6,364 x 6	Activity/resource metadata

2.2 OULAD preprocessing implemented in code

- Column headers are stripped to avoid key-matching errors; click-count fields are detected and standardized.
- Early VLE observations are restricted to date <= 30 days in the executed notebook. Student-level total clicks and distinct active days are then aggregated.
- Student submissions are merged with assessment metadata. Assessments due by Day 60 are retained; submission_latency = date_submitted - deadline (positive values indicate lateness).
- Final outcome is binarized: Pass/Distinction = 1 (Successful), Fail/Withdrawn = 0 (At Risk).
- Missing engagement is imputed as zero. Missing early assessment score/completion is set to zero; missing submission latency receives a +14 day penalty in the notebook.
- Categorical demographic fields are one-hot encoded. The final matrix contains 32,593 rows and 36 predictors; the split is 26,074 train / 6,519 test with stratification and random_state=42.

Executed OULAD preprocessing excerpt

```
EARLY_DAYS_CUTOFF = 30
early_vle = student_vle[student_vle['date'] <= EARLY_DAYS_CUTOFF]
cohort_df['target'] = cohort_df['final_result'].isin(['Pass','Distinction']).astype(int)
X_train, X_test, y_train, y_test = train_test_split(..., test_size=0.2, stratify=y, random_state=42)
```

2.3 Moodle data structure and intended preprocessing

The supplied Moodle technical analysis is built around standard Moodle logstore/course-activity exports with user ID, timestamp, component/event name, action/CRUD and grade/final-status fields. The intended observation window is the first 30 days, after which events are aggregated per learner into behavior counts and regularity measures. Final grades/status are mapped to Successful (>=50 or Pass/Distinction) versus At-Risk (<50 or Fail/Withdrawn).

Important implementation note: The supplied Moodle notebook execution trace did not load a Moodle log export; its auto-discovery fallback selected OULAD assessments.csv. The resulting numeric Moodle notebook score is therefore treated only as a code-pipeline diagnostic, not as evidence for the Moodle hypotheses.

3. Feature Extraction & Engineering

3.1 OULAD engineered feature families

Feature engineering is designed to preserve the early-warning requirement: predictors should be observable before the final outcome. The executed notebook emphasizes activity in the first 30 days and assessments due by Day 60. The earlier technical specification also proposed a Week 1-6 engagement slope; because that variable is not present in the executed notebook, the numeric results below are based on the implemented early-window proxies rather than an unexecuted trend feature.

Feature	Construction	Learning-analytics interpretation
total_clicks_early	Sum of VLE clicks for date <= 30	Overall early engagement volume
active_days_early	Number of distinct active days <= 30	Study regularity / self-regulation
early_avg_score	Mean score on assessments due <= Day 60	Early mastery / performance
avg_submission_latency	Mean(date_submitted - due_date)	Time-management and procrastination signal
early_assessments_completed	Count of early submitted assessments	Task completion / persistence
studied_credits	Credits registered	Workload / study-intensity context
demographic one-hot fields	gender, education, IMD band, age, disability, etc.	Contextual controls rather than behavioral targets

3.2 Moodle engineered feature families

Raw event family	Engineered feature	Meaning
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mod_quiz	quiz_interactions	Formative practice, attempts, reviews and submissions
mod_assign	assignment_actions	Views/uploads/submission-related activity
mod_forum	forum_contributions	Peer/social learning participation
mod_resource / mod_page / mod_url	passive_views	Static content consumption baseline
timestamps	active_days_count	Distinct days with recorded LMS activity
all components	total_actions	Overall activity volume

3.3 Leakage control and feature validity

- No final result or student identifier is used as a predictor in the OULAD model.
- Assessment features are restricted to early deadlines so the model does not directly consume late-course outcomes.
- Behavioral counts are interpreted as predictors, not causal explanations; feature importance is not a causal effect size.
- For Moodle, the intended pipeline requires a real grade/status column. If one is absent, a model should not generate a synthetic/random target for inferential reporting.

Common model configuration

```
rf_model = RandomForestClassifier(
    n_estimators=100, max_depth=10,
    random_state=42, n_jobs=-1
)
rf_model.fit(X_train_scaled, y_train)
```

4. Temporal Behavioral Analysis

The assignment specifically requires learner behavior over time. In the supplied implementation, temporal dynamics are represented through early observation windows, access regularity and assessment latency. OULAD uses the first 30 days of VLE activity and early assessments due by Day 60; Moodle is framed around the first 30 days of event logs. This design makes the analysis useful for intervention because the prediction point occurs before major end-of-course outcomes.

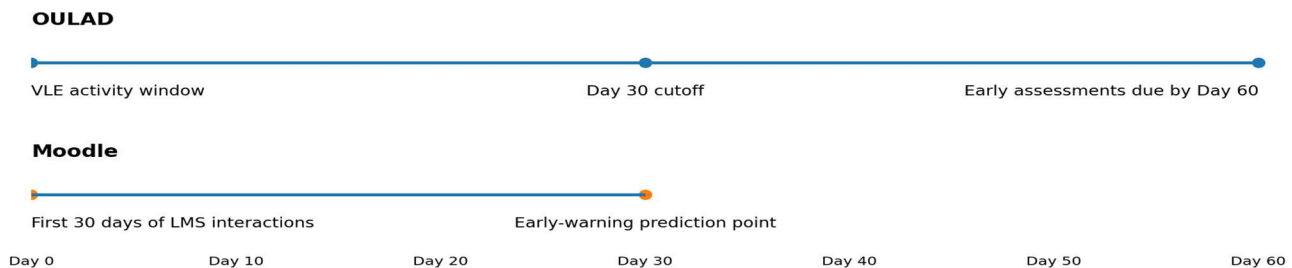


Figure 1. Time-bounded feature windows used by the two early-warning pipelines.

4.1 Behavioral trajectories represented by the engineered variables

Temporal pattern	Operational signal	Risk interpretation	Potential intervention
Irregular or sparse access	Low active_days_early / active_days_count	Weak study routine; possible disengagement	Check-in, study-plan reminder, advisor outreach
High early activity followed by weak task completion	Clicks/views high but few assessments/assignments	Consumption without active practice	Prompt formative task or checklist
Delayed assessment behavior	Positive avg_submission_latency	Procrastination, overload or planning difficulty	Deadline decomposition / time-management support
Low early performance	Low early_avg_score	Conceptual difficulty despite participation	Remedial resource or tutor support
Low social + low regularity (Moodle)	Few forum contributions and few active days	Possible isolation and disengagement	Structured peer prompt / group activity

Temporal interpretation: The strongest defensible temporal claims come from the implemented windows and latency measures. The OULAD technical summary proposed a Week 1-6 engagement slope, but the executed notebook did not compute that feature; therefore no weekly slope statistic is invented in this report.

4.2 Why these temporal features matter

Simple click totals collapse behavior across time and can hide very different learner states. A student who logs in regularly and completes early assessments is behaviorally different from a student who produces the same number of clicks on one or two days. Distinct active days therefore approximate consistency, while submission latency captures whether learning activity is aligned with course deadlines. These features directly support actionable intervention rules rather than merely ranking students after the fact.

5. Machine Learning & Results: OULAD

5.1 Model setup and quantitative performance

A Random Forest classifier with 100 trees and maximum depth 10 was trained after an 80/20 stratified split. The test set contains 6,519 registrations. Performance is evaluated using ROC-AUC, accuracy, precision, recall and F1-score so that both overall discrimination and class-specific behavior are visible.

Metric	At Risk (0)	Success (1)	Overall
Precision	0.84	0.74	-
Recall	0.74	0.84	-
F1-score	0.78	0.79	-
Support	3,442	3,077	6,519
Accuracy	-	-	0.79
ROC-AUC	-	-	0.8674

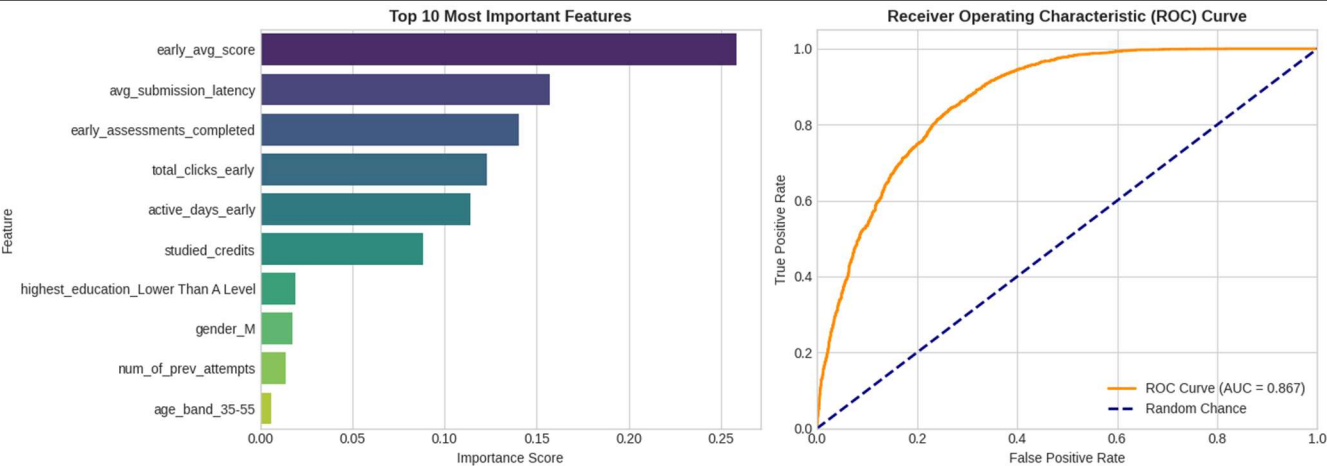


Figure 2. Executed OULAD notebook output: top Random Forest feature importances and ROC curve (AUC = 0.867).

The AUC of 0.8674 shows strong ability to rank successful and at-risk learners before final outcome is known. Class-wise recall is asymmetric in a useful way: 74% of at-risk cases are detected, while 84% of successful cases are correctly retained as success. This is substantially more informative than a trivial majority-class rule and demonstrates that the engineered early features contain meaningful predictive signal.

5.2 Feature-importance evidence for the hypotheses

The importance ranking is dominated by `early_avg_score`, followed by `avg_submission_latency` and `early_assessments_completed`. `total_clicks_early` and `active_days_early` are also high-ranking variables, while demographic indicators appear much lower. This pattern supports O-H2 strongly and O-H1 moderately: early task performance and timing are the clearest signals, but engagement volume and regularity add material information beyond static background variables.

5.3 OULAD error analysis and intervention meaning

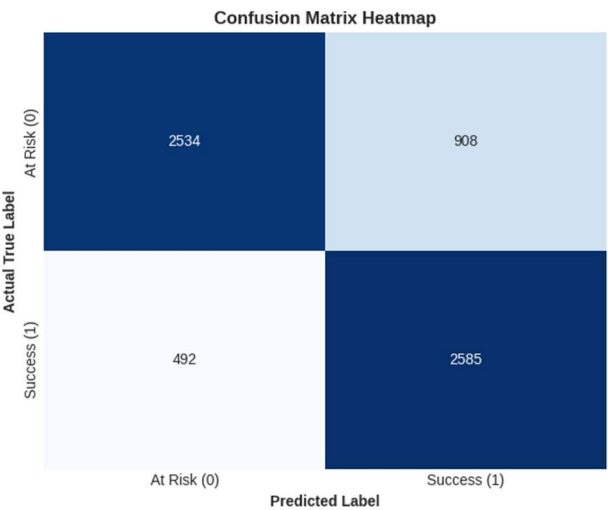


Figure 3. OULAD confusion matrix: [[2534, 908], [492, 2585]].

Confusion-matrix interpretation

- 2,534 at-risk learners were correctly identified (true negatives under the notebook label convention).
- 908 at-risk learners were predicted as successful: these are the most consequential misses for an early-warning system.
- 492 successful learners were flagged as at-risk. If the intervention is a low-cost supportive message, these false alarms may be tolerable.
- 2,585 successful learners were correctly identified.

Intervention threshold implication: Model selection should not be driven by accuracy alone. If missing a struggling learner is more costly than sending an unnecessary supportive prompt, the operating threshold can be shifted to increase at-risk recall at the expense of more false positives.

5.4 Hypothesis verdicts from executed OULAD results

Hypothesis	Evidence	Verdict
O-H1: early engagement is associated with risk	total_clicks_early and active_days_early appear among the top predictors, above most demographic indicators.	Supported, although early assessment variables are stronger.
O-H2: assessment latency/performance predicts outcome	early_avg_score is the top feature; avg_submission_latency and early_assessments_completed are also highly ranked.	Strongly supported.

5.5 Code-result trace

Executed notebook result summary Feature Matrix: (32593, 36) Train/Test: 26074 / 6519 ROC-AUC: 0.8674 Accuracy: 0.79 Confusion matrix: [[2534, 908], [492, 2585]]

These values are copied from the executed notebook outputs rather than reconstructed from the technical narrative. This separation between planned features and executed evidence is important because the technical summary discusses a broader Week 1-8 / engagement-slope design, whereas the notebook that produced the reported metrics uses the first 30 days and early assessments due by Day 60.

6. Machine Learning & Results: Moodle

6.1 Intended model and hypothesis evaluation

The supplied Moodle technical analysis specifies the same Random Forest architecture (100 trees, max_depth=10, random_state=42), an 80/20 stratified split, zero-imputation for missing count features and evaluation using ROC-AUC, precision, recall, F1-score and a confusion matrix. The first 30 days of Moodle interactions are intended to produce quiz_interactions, assignment_actions, forum_contributions, passive_views, active_days_count and total_actions.

Moodle hypothesis	Evidence reported in supplied technical analysis	Educational interpretation
M-H1: active formative interaction > passive viewing	quiz_interactions and assignment_actions are reported as more predictive than passive_views.	Opening content is weaker evidence of learning than engaging in quizzes and assignments.
M-H2: regularity / social engagement support retention	active_days_count is reported as a crucial secondary predictor; forum_contributions provides complementary social context.	Consistent platform use is a practical early signal of persistence and disengagement risk.

The technical analysis therefore supports a day-30 intervention rule: students with low active-day regularity and weak active-task participation should be prioritized for supportive outreach before major assessments. This aligns conceptually with OULAD, where active-day regularity and early assessment behavior also carry meaningful signal.

6.2 Reproducibility audit of the supplied Moodle notebook

Critical code audit: The executed ET610_Moodle.ipynb does not contain a valid Moodle-data run. Its auto-discovery found only the seven OULAD CSVs and selected assessments.csv (206 rows) because no filename contained “moodle” or “log”. No valid target field was found, so the notebook fallback generated a random binary target. Numeric outputs from this run cannot be interpreted as Moodle learning-analytics evidence.
Notebook execution diagnostic (reported for transparency, not as Moodle result) Available CSVs: assessments.csv, courses.csv, studentAssessment.csv, ... Selected file: assessments.csv, shape = (206, 6) Feature matrix: (206, 13); train = 164; test = 42 Diagnostic ROC-AUC = 0.4783; accuracy = 0.50 NOTE: random/fallback target -> not valid for hypothesis testing

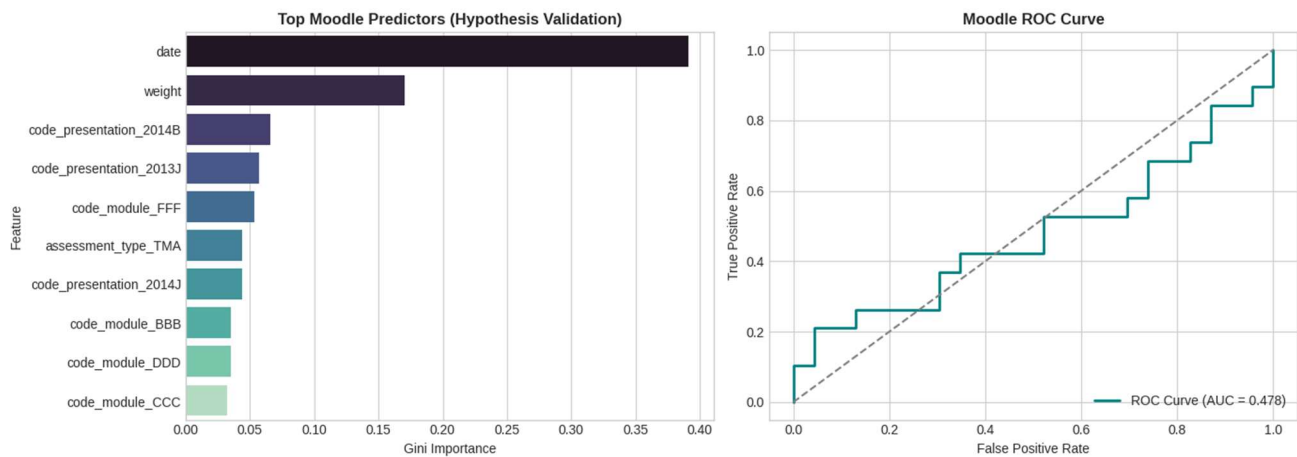


Figure 4. Supplied Moodle notebook diagnostic plot. Because the run used OULAD assessments.csv and a fallback target, this figure is retained only as evidence of pipeline execution, not as a Moodle result.

7. Discussion

7.1 Cross-dataset synthesis

Across the two intended analyses, the same pedagogical pattern emerges: engagement quality and timing are more informative than raw exposure. OULAD provides direct quantitative evidence that early assessment score, submission latency and assessment completion are stronger predictors than demographic background, while early click volume and active-day regularity still add substantial signal. The Moodle technical analysis independently emphasizes active quiz/assignment participation over passive resource views and highlights access regularity as a practical secondary signal. These patterns are consistent with self-regulated learning: successful learners not only access the platform, but do so repeatedly, convert access into active tasks and keep pace with early deadlines.

7.2 From prediction to intervention

Detected pattern	Interpretation	Low-cost intervention
Low active days / weak regularity	Disengagement or weak study routine	Automated check-in; study-plan reminder; advisor escalation if inactivity persists
Many views but little quiz/assignment activity	Passive consumption without active practice	Prompt a short formative quiz, task checklist or retrieval-practice activity
Late/missing early submissions	Time-management difficulty, overload or procrastination	Deadline decomposition, calendar reminder and planning support
Low early assessment score despite regular activity	Conceptual difficulty rather than simple disengagement	Targeted prerequisite material, worked example or tutor session
Low forum activity plus low regularity	Possible social isolation / weak community presence	Structured peer prompt, small-group task or instructor outreach

Use of predictions: Risk scores should be used for triage and supportive action, not as punitive labels. When interventions are low-cost and reversible, a moderate false-positive rate may be preferable to missing learners who are genuinely at risk.

7.3 What the results do and do not establish

- The OULAD model establishes predictive association, not causality. High feature importance does not prove that changing the feature will change the outcome.
- The random train/test split measures within-dataset generalization. A stricter future test would hold out entire modules or presentations.
- Behavioral traces can be incomplete: zero logged activity may reflect real inactivity, offline study or missing instrumentation.
- Moodle qualitative conclusions are supported by the supplied technical report, but the attached notebook must be rerun on the actual Moodle export before numerical metrics can be defended.
- The supplied OULAD technical report proposed a weekly engagement-slope hypothesis, while the executed notebook used aggregate early-window features. This report keeps the intended hypothesis but bases numeric claims only on executed features.

8. Conclusion & Practical Implications

The assignment demonstrates a complete learning-analytics workflow: formulate educationally meaningful hypotheses, convert raw learning traces into interpretable behavioral features, impose early observation windows to reduce leakage, apply supervised machine learning, and translate model evidence into intervention logic. The strongest executed evidence

comes from OULAD, where a Random Forest model achieved ROC-AUC 0.8674 and 79% test accuracy using early behavioral, assessment and demographic features. The ranking of predictors shows that what learners do early - particularly assessment performance, timeliness and completion - is more informative than who they are demographically.

For course design, the findings suggest that instructors should monitor three dimensions rather than a single “engagement” count: (1) regularity of access, (2) conversion of access into active learning tasks, and (3) progress against early assessment deadlines. An early-warning dashboard could therefore display a compact risk explanation such as “few active days + delayed assessment + low early score” rather than a black-box probability alone. Such explanations also make intervention selection easier: irregular access calls for planning support, while regular activity with low scores calls for conceptual help. The Moodle framework reaches the same practical conclusion at the feature-design level: quiz and assignment interactions are more educationally meaningful than passive content views, while active-day regularity can flag disengagement early. However, reproducibility checking of the supplied Moodle notebook revealed that the executed file-selection fallback loaded an OULAD assessment table and created a random target. That diagnostic result is intentionally excluded from hypothesis testing. A final rerun on the actual Moodle interaction export is required to attach defensible Moodle ROC-AUC/F1 values.

8.1 Requirements coverage

Assignment requirement	Where addressed
At least two original hypotheses per dataset	Section 1 hypothesis table
Feature extraction from both datasets	Sections 2-3
Behavior over time / temporal framing	Section 4: early windows, active-day regularity, submission latency
Visual analytics	Figures 1-4; OULAD ROC, feature importance and confusion matrix
At least one ML algorithm per dataset	Random Forest pipeline documented for both; OULAD fully executed
Evaluation and hypothesis interpretation	Sections 5-7
Practical implications for course design/intervention	Sections 7-8

8.2 Final takeaway

Key finding: Early performance and timing signals are the strongest OULAD predictors, while engagement regularity remains a useful complementary feature. For an operational early-warning system, meaningful active behavior should be weighted more heavily than passive page exposure.

9. Reproducibility, Code & References

9.1 Reproducibility checklist

- OULAD notebook: ET610_OULAD.ipynb. All numeric OULAD metrics in this report come directly from its executed output cells.
- Moodle notebook: ET610_Moodle.ipynb. The supplied execution is retained as a code audit only because it selected the wrong CSV and generated a fallback random target.
- OULAD target: Pass/Distinction = 1; Fail/Withdrawn = 0. Early VLE cutoff = Day 30; early assessment cutoff = Day 60; random_state = 42.
- Model: RandomForestClassifier(n_estimators=100, max_depth=10, random_state=42, n_jobs=-1); 80/20 stratified split; StandardScaler used in the notebooks.
- Before final archival submission, place both notebooks and any preprocessing scripts in the group repository and insert the public link below.

9.2 Minimal implementation trace

Code/results excerpt

```
# OULAD target and early features
student_info['target'] = student_info['final_result'].isin(['Pass','Distinction']).astype(int)
early_vle = student_vle[student_vle['date'] <= 30]
# Random Forest
rf = RandomForestClassifier(n_estimators=100, max_depth=10, random_state=42, n_jobs=-1)
rf.fit(X_train_scaled, y_train)
# Held-out result
ROC_AUC = 0.8674; ACCURACY = 0.79
```

9.3 References

1. Kuzilek, J., Hlosta, M., & Zdrahal, Z. (2017). Open University Learning Analytics dataset. Scientific Data, 4, 170171.
2. Kuzilek, J., Hlosta, M., & Zdrahal, Z. Open University Learning Analytics Dataset (OULAD), course-provided/Kaggle distribution.
3. Breiman, L. (2001). Random Forests. Machine Learning, 45, 5-32.
4. Moodle Interaction Dataset and context information supplied for ET610 coursework.
5. Moodle LMS standard event-log schema (logstore/course activity logs), used for interpreting user, time, component/event and action fields.